

# CH16 Regression Diagnostic

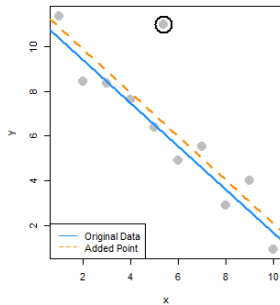
# 회귀진단

- (1) 오차항의 가정 검토
- (2) 적절한 모형의 선택
- (3) 독립변수들간의 상관관계 검토
- (4) 지렛대점(leverage point)의 검출
- (5) 이상치(outlier) 확인
- (6) 영향점(influential observation)의 검출

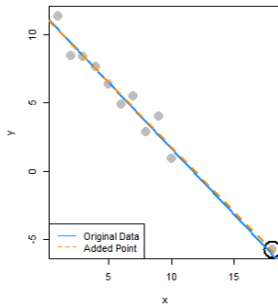
# 회귀진단

- leverage vs. outlier vs. influence point

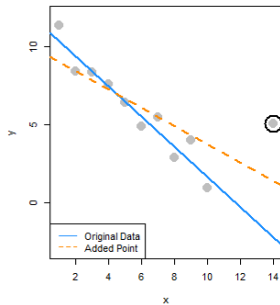
Low Leverage, Large Residual, Small Influence



High Leverage, Small Residual, Small Influence



High Leverage, Large Residual, Large Influence



# Hat matrix

- 추정된 회귀직선

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H} \mathbf{y}$$

- $H$  : hat matrix,  $n \times n$  matrix
- $\text{Var}(\hat{\mathbf{y}}) = \sigma^2 H$ ,  $\text{Var}(\mathbf{e}) = (\mathbf{I}_n - H)\sigma^2$
- for  $i, j = 1, \dots, n$

$$h_{ij} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_j$$

where  $\mathbf{x}_i = (1, x_{i1}, x_{i2}, \dots, x_{ip})$

- $h_{ii}$  :  $H$ 의 대각 원소 (diagonal elements)

$$H = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

- $HH = H$ , idempotent matrix,  $\text{rank}(H) = p + 1$ ,

$$\text{tr}(H) = \sum_{i=1}^n h_{ii} = p + 1$$

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$$

- $H$  : positive definite (양반정치행렬)

$$0 \leq h_{ii} < 1, \quad -1/2 \leq h_{ij} \leq 1/2$$

# Hat matrix

- $p = 1$

$$\begin{aligned}h_{ij} &= \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_j \\&= \begin{pmatrix} 1 & x_i \end{pmatrix} \begin{pmatrix} \frac{\sum x_i^2}{nS_{(xx)}} & \frac{-\bar{x}}{S_{(xx)}} \\ \frac{-\bar{x}}{S_{(xx)}} & \frac{1}{S_{(xx)}} \end{pmatrix} \begin{pmatrix} 1 \\ x_j \end{pmatrix} = \frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{(xx)}} \\h_{ii} &= \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{(xx)}}\end{aligned}$$

# Hat matrix

- $p > 1$

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_i - \bar{\mathbf{x}})^\top (\mathcal{X}^\top \mathcal{X})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

where  $\mathbf{x}_i^\top = (x_{i1}, x_{i2}, \dots, x_{ip})$ ,  $\bar{\mathbf{x}}^\top = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)$ ,

$$\bar{x}_j = \sum_{i=1}^n x_{ij}/n,$$

$$\mathcal{X} = \begin{pmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \cdots & x_{1p} - \bar{x}_p \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \cdots & x_{2p} - \bar{x}_p \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \cdots & x_{np} - \bar{x}_p \end{pmatrix}$$

## Leverage point

- $h_{ii} > 2\bar{h}$  이면,  $i$  번째 관측치가 leverage point로 고려 가능

$$\bar{h} = \frac{1}{n} \sum_{i=1}^n h_{ii} = \frac{p+1}{n}$$



$$\begin{aligned}\mathbf{e} &= \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{y} - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \\ &= [\mathbf{I}_n - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T] \mathbf{y} = (\mathbf{I}_n - H) \mathbf{y}\end{aligned}$$

- $E(\mathbf{e}) = \mathbf{0}_n$ ,  $\text{Var}(\mathbf{e}) = (\mathbf{I}_n - H)\sigma^2$
- $E(e_i) = 0$ ,  $\text{Var}(e_i) = (1 - h_{ii})\sigma^2$ ,  $\text{Cov}(e_i, e_j) = -h_{ij}\sigma^2$

$$\text{Corr}(e_i, e_j) = \rho_{ij} = \frac{-h_{ij}}{\sqrt{(1 - h_{ii})(1 - h_{jj})}}$$

## Standardized residual

- Since  $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \mathbf{I}_n\sigma^2)$ ,  $\mathbf{e} \sim N(\mathbf{0}_n, (\mathbf{I}_n - H)\sigma^2)$  and

$$e_i \sim N(0, (1 - h_{ii})\sigma^2)$$

- (내적) 표준화잔차 ((internally) standardized residual)

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}, \quad \hat{\sigma}^2 = MSE$$

## Studentized residual

- (외적) 스튜던트화 잔차 ((externally) studentized residual)

$$r_i^* = \frac{e_i}{\hat{\sigma}_{(i)} \sqrt{1 - h_{ii}}}$$

단,  $\hat{\sigma}_{(i)}$  :  $i$  번째 측정값  $y_i$  를 제외하고 얻어진  $\hat{\sigma}$

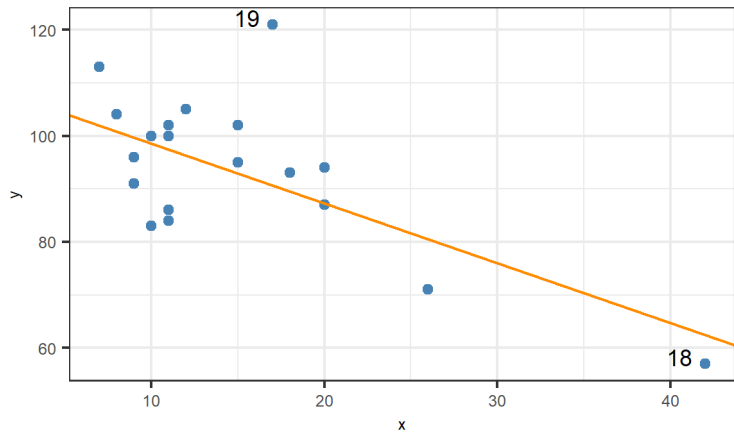
$$\hat{\sigma}_{(i)}^2 = \left[ (n - p - 1) \hat{\sigma}^2 - \frac{e_i^2}{1 - h_{ii}} \right] / (n - p - 2)$$

- $|r_i^*| \geq t_{\alpha/2}(n - p - 2)$  이면, 유의수준  $\alpha$ 에서,  $y_i$  를 이상점이라고 판정

# Example

| 실험번호 | $x$ | $y$ | 실험번호 | $x$ | $y$ | 실험번호 | $x$ | $y$ |
|------|-----|-----|------|-----|-----|------|-----|-----|
| 1    | 15  | 95  | 8    | 11  | 100 | 15   | 11  | 102 |
| 2    | 26  | 71  | 9    | 8   | 104 | 16   | 10  | 100 |
| 3    | 10  | 83  | 10   | 20  | 94  | 17   | 12  | 105 |
| 4    | 9   | 91  | 11   | 7   | 113 | 18   | 42  | 57  |
| 5    | 15  | 102 | 12   | 9   | 96  | 19   | 17  | 121 |
| 6    | 20  | 87  | 13   | 10  | 83  | 20   | 11  | 86  |
| 7    | 18  | 93  | 14   | 11  | 84  | 21   | 10  | 100 |

# Example



## Example

- 회귀직선  $\hat{y} = 109.874 - 1.127x$
- 분산분석표

| 요인 | 제곱합      | 자유도 | 평균제곱     | $F_0$ |
|----|----------|-----|----------|-------|
| 회귀 | 1,604.08 | 1   | 1,604.08 | 12.20 |
| 잔차 | 2,308.59 | 19  | 121.505  |       |
| 계  | 3,912.67 | 20  |          |       |

- $F_0 = 13.20 > F_{0.05}(1, 19) = 4.38$  이므로 회귀직선은 유의
- $\hat{\sigma} = \sqrt{121.505} = 11.0229$

# Example

Table: 지렛대점과 이상점을 찾는 측도( $h_{ii}$ ,  $r_i$ 와  $r_i^*$ )

| 실험번호 | $e_i$    | $h_{ii}$ | $r_i$   | $r_i^*$ |
|------|----------|----------|---------|---------|
| ·    | ·        | ·        | ·       | ·       |
| ·    | ·        | ·        | ·       | ·       |
| 8    | 2.5230   | 0.0567   | 0.2357  | 0.2297  |
| 9    | 3.1421   | 0.0799   | 0.2972  | 0.2899  |
| 10   | 6.6666   | 0.0726   | 0.6280  | 0.6177  |
| 11   | 11.0151  | 0.0908   | 1.0480  | 1.0508  |
| 12   | -3.7309  | 0.0705   | -0.3511 | -0.3428 |
| 13   | -15.6040 | 0.0628   | -1.4623 | -1.5108 |
| 14   | -13.4770 | 0.0567   | -1.2588 | -1.2798 |
| 15   | 4.5230   | 0.0567   | 0.4225  | 0.4132  |
| 16   | 1.3961   | 0.0628   | 0.1308  | 0.1274  |
| 17   | 8.6500   | 0.0521   | 0.8060  | 0.7983  |
| 18   | -5.5403  | 0.6516   | -0.8515 | -0.8451 |
| 19   | 30.2850  | 0.0531   | 2.8234  | 3.6070  |
| 20   | -11.4770 | 0.0567   | -1.0720 | -1.0765 |
| 21   | 1.3961   | 0.0628   | 0.1308  | 0.1274  |

## Example

- $\bar{h} = \frac{p+1}{n} = \frac{2}{21} = 0.0952$
- $2\bar{h} = 0.1905, \quad 3\bar{h} = 0.2857$
- $t_{0.025}(n-p-2) = t_{0.025}(18) = 2.101$
- 18번째 데이터 : leverage point
- 19번째 데이터 : outlier



# 영향점을 검출하는 방법

## (1) DFFITS (Difference if Fits)

$$\text{DFFITS}(i) = \frac{\hat{y}_i - \tilde{y}_i(i)}{\hat{\sigma}_{(i)} \sqrt{h_{ii}}}$$

- ▶  $\tilde{y}_i(i)$  :  $i$  번째 데이터를 제외시키고  $n - 1$  개 데이터에서 얻은 예측값
- ▶  $|\text{DFFITS}(i)| \geq 2\sqrt{\frac{p+1}{n-p-1}} \Rightarrow \text{영향점}$

# 영향점을 검출하는 방법

## (2) Cook's Distance

$$D(i) = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_j(i))^2}{(p+1)\hat{\sigma}^2} = \frac{(\hat{\beta} - \hat{\beta}(i))^T \mathbf{X}^T \mathbf{X} (\hat{\beta} - \hat{\beta}(i))}{(p+1)\hat{\sigma}^2}$$

- ▶  $\hat{\beta}(i)$  :  $i$  번째 관측치를 제외시키고  $n - 1$  개의 관측값에서 구한  $\beta$ 의 최소제곱추정량

$$D(i) = \frac{h_{ii}}{1 - h_{ii}} \cdot \frac{1}{p+1} = \frac{r_i^2}{p+1} \cdot \frac{h_{ii}e_i^2}{\hat{\sigma}^2(1 - h_{ii})^2}$$

- ▶  $D(i) \geq F_{0.5}(p+1, n-p-1)$  이면 영향점으로 의심

# 영향점을 검출하는 방법

## (3) Andrews-Pregibon의 통계량

- ▶ 행렬  $\mathbf{X}$ 와 벡터  $\mathbf{y}$ 를 같이 고려

$$AP(i) = \frac{|\mathbf{X}^*(i)^\top \mathbf{X}^*(i)|}{|\mathbf{X}^{*\top} \mathbf{X}^*|}$$

- ▶  $\mathbf{X}^* = (\mathbf{X}, \mathbf{y})$ ,  $\mathbf{X}^*(i) = (\mathbf{X}(i), \mathbf{y}(i))$
- ▶  $\mathbf{X}^*(i)$  :  $\mathbf{X}^*$  행렬에서  $i$  번째 행을 제거한 것
- ▶ 가장 작은  $AP(i)$ 의 값을 영향점으로 간주

## (4) COVRATIO

$$\text{COVRATIO}(i) = \frac{1}{\left[1 + \frac{(r_i^*)^2 - 1}{n-p-1}\right]^{p+1} (1 - h_{ii})}$$

- ▶ COVRATIO의 값이 1에 가까우면  $y_i$ 는 별로 영향을 주지 못함
- ▶ 1에서 멀어질수록 영향을 크게 주고 있음
- ▶  $|\text{COVRATIO}(i) - 1| \geq 3(p+1)/n$  이면  $i$ 번째 관측치를 영향을 크게 주는 측정값으로 볼 수 있음

# 영향점을 검출하는 방법

## (5) FVARATIO

$$\begin{aligned}\text{FVARATIO}(i) &= \frac{h_{ii}\hat{\sigma}^2(i)/(1-h_{ii})}{h_{ii}\hat{\sigma}^2} = \frac{\hat{\sigma}_{(i)}^2}{\hat{\sigma}^2(1-h_{ii})} \\ &= \frac{e_i^2}{(r_i^*)^2(1-h_{ii})^2\hat{\sigma}^2}\end{aligned}$$

- ▷  $\text{FVARATIO}(i) \leq 1 - \frac{3}{n}$  이거나  $\text{FVARATIO}(i) \geq 1 + \frac{2p+3}{n}$  이면  $i$  번째 관측치를 영향을 크게 주는 측정값으로 제안

# 지렛대점, 이상점과 영향점을 찾는 데 사용되는 척도

| 지렛대점                                                                                                          | 이상점                                                                                                                                                                                  | 영향점                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. 행렬 H의 대각선원소</p> $h_{ii}$ <p>2. Mahalanobis의 거리</p> $M(i) = (n - 1) \left( h_{ii} - \frac{1}{n} \right)$ | <p>1. 표준화잔차</p> $r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$ <p>2. 스튜던트화 잔차</p> $r_i^* = \frac{e_i}{s(i)\sqrt{1 - h_{ii}}} = r_i \left( \frac{n - p - 2}{n - p - 1 - r_i^2} \right)^{1/2}$ | <p>1. DFFITS(i) = <math>\left( \frac{h_{ii}}{1 - h_{ii}} \right)^{1/2} r_i^*</math></p> <p>2. Cook의 통계량</p> $D(i) = \frac{h_{ii}}{(p + 1)(1 - h_{ii})} \cdot r_i^2$ <p>3. Andrews-Pregibon의 통계량</p> $AP(i) = 1 - h_{ii} - \frac{e_i^2}{(n - p - 1)s^2}$ <p>4. COVRATIO(i)</p> $= \frac{1}{\left[ 1 + \frac{(r_i^*)^2 - 1}{n - p - 1} \right]^{p+1}} \cdot (1 - h_{ii})$ <p>5. FVARATIO(i)</p> $= \frac{e_i^2}{(r_i^*)^2(1 - h_{ii})^2 s^2}$ |

# Example

| 실험번호 | DFFITS( $i$ ) | D( $i$ ) | M( $i$ ) | AP( $i$ ) | COVRATIO( $i$ ) | FVARATIO( $i$ ) |
|------|---------------|----------|----------|-----------|-----------------|-----------------|
| ⋮    | ⋮             | ⋮        | ⋮        | ⋮         | ⋮               | ⋮               |
| 7    | 0.0772        | 0.0031   | 0.2074   | 0.9370    | 1.1702          | 1.1145          |
| 8    | 0.0563        | 0.0017   | 0.1810   | 0.9406    | 1.1742          | 1.1157          |
| 9    | 0.0854        | 0.0038   | 0.6448   | 0.9159    | 1.1997          | 1.1418          |
| 10   | 0.1728        | 0.0154   | 0.5000   | 0.9081    | 1.1521          | 1.1146          |
| 11   | 0.3320        | 0.0548   | 0.8027   | 0.8567    | 1.0878          | 1.0938          |
| 12   | -0.0944       | 0.0047   | 0.4585   | 0.9234    | 1.1833          | 1.1283          |
| 13   | -0.3911       | 0.0717   | 0.3039   | 0.8317    | 0.9363          | 0.9996          |
| 14   | -0.3137       | 0.0416   | 0.1810   | 0.8647    | 0.9923          | 1.0256          |
| 15   | 0.1013        | 0.0054   | 0.1810   | 0.9345    | 1.1590          | 1.1085          |
| 16   | 0.0330        | 0.0006   | 0.3039   | 0.9363    | 1.1867          | 1.1253          |
| 17   | 0.1972        | 0.0179   | 0.0898   | 0.9155    | 1.0964          | 1.0755          |
| 18   | -1.1558       | 0.6781   | 12.0498  | 0.3351    | 2.9587          | 2.9142          |
| 19   | 0.8537        | 0.2233   | 0.1086   | 0.5497    | 0.3964          | 0.6470          |
| 20   | 0.2638        | 0.0345   | 0.1810   | 0.8863    | 1.0426          | 1.0513          |
| 21   | 0.0330        | 0.0006   | 0.3039   | 0.9363    | 1.1867          | 1.1253          |